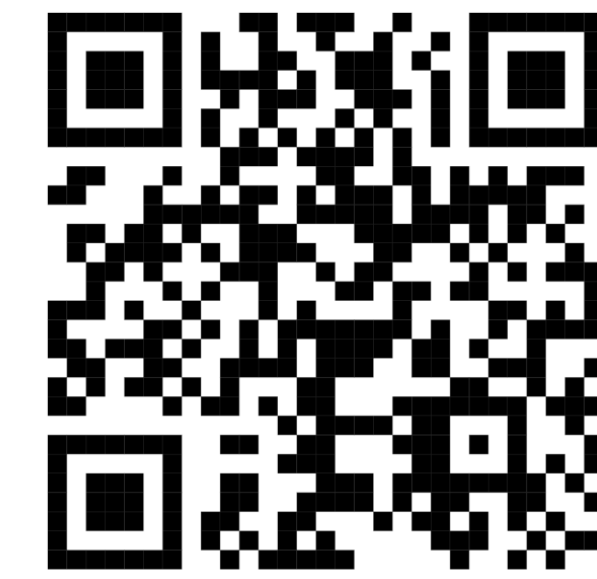


YUPU: A Benchmark and Multimodal Diffusion Model for ALS Point Cloud Upsampling

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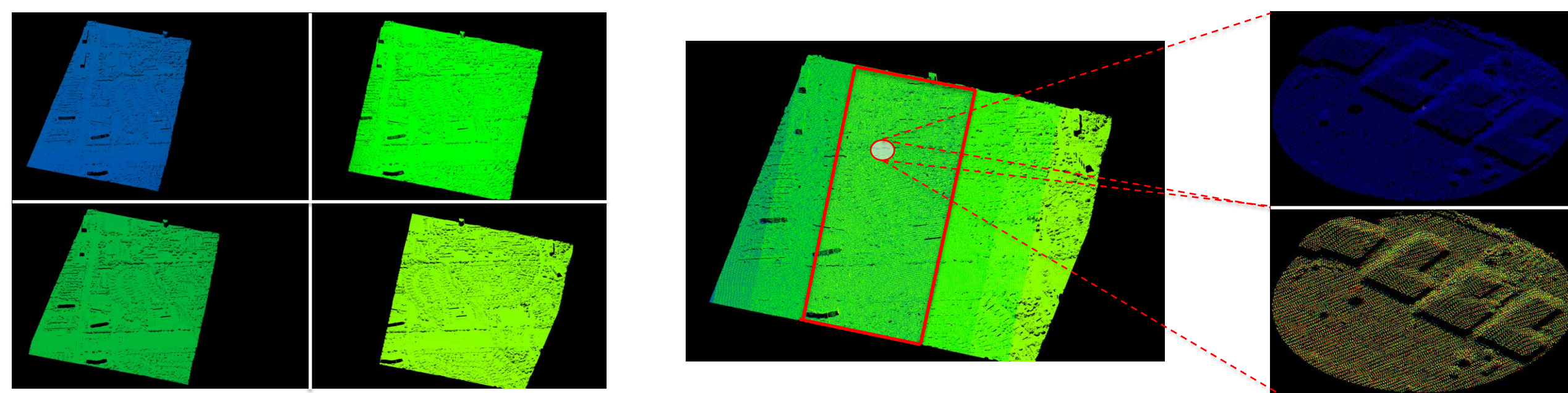
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1. Motivation

- ❑ Dense ALS acquisition is costly.
- ❑ There is lack of existing benchmark dataset for ALS point cloud upsampling task.

2. YUPU Benchmark



(a) Point Cloud with individual Scan IDs

(b) Point Cloud with All Scan IDs

(c) Sampled Scenes

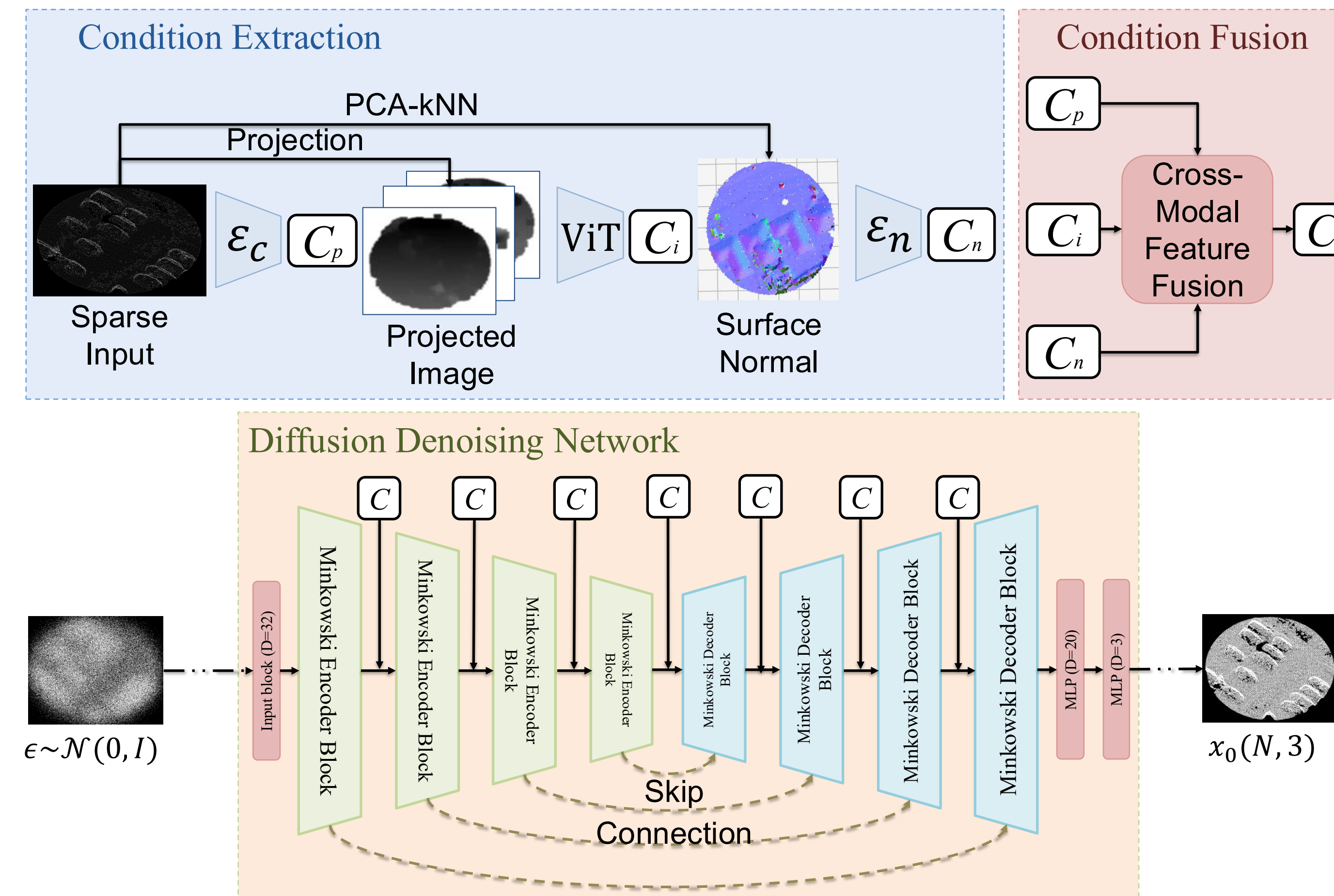
Property	Description
Spatial coverage	5.4 km ²
Survey areas	13
Number of scenes	1300
Scene radius	30m
Sparse input	12,000 points
Dense reference	x2, x3, x4
Point attributes	XYZ, intensity, return number, ScanID

❑ Synthetic vs Acquisition

Training input	Test input	CD↓	EMD↓	JSD-3D↓	F1↑
Synthetic	Synthetic	0.415	2.517	0.646	0.844
Synthetic	Acquisition-derived	0.437	2.623	0.683	0.825
Acquisition-derived	Acquisition-derived	0.414	2.651	0.655	0.883

- ❑ Existing diffusion models have limited conditioning.

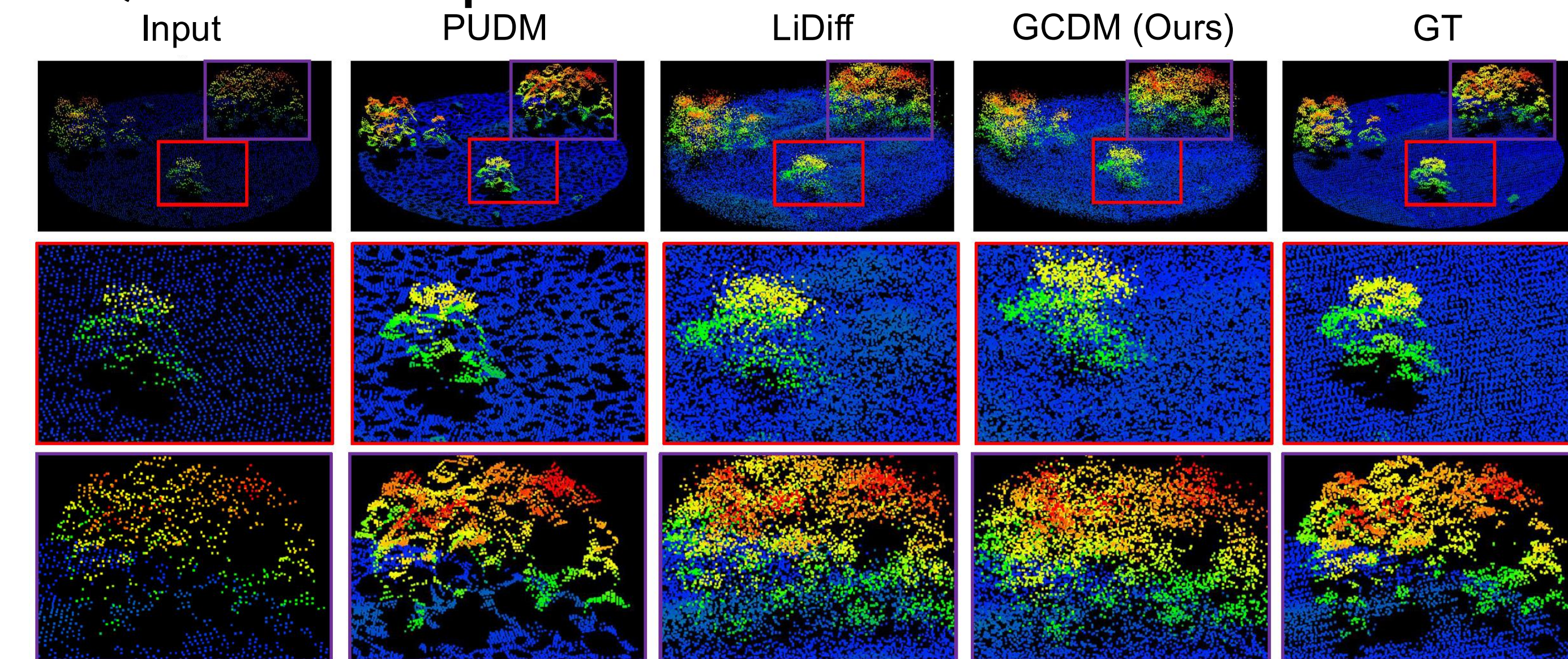
3. GCDM



- ❑ Point condition C_p : Preserves observed spatial information.
- ❑ Appearance condition C_i : Multi-view features provide broader structure context.
- ❑ Geometry condition C_n : Surface normal encode local orientation and curvature
- ❑ GCDM retains the LiDiff diffusion backbone and objective, isolating the effect of improved conditioning.

4. Experimental Results

❑ Qualitative Comparison



❑ Quantitative Comparison

Model	Geometric fidelity			JSD Distribution		Local	Completion IoU		
	CD↓	HD↓	EMD↓	BEV↓	3D↓	F1↑	0.5 m↑	1 m↑	2 m↑
LiDiff [17]	0.606	8.975	3.001	0.439	0.703	0.725	0.144	0.310	0.455
PUDM [19]	1.228	6.570	7.130	0.467	0.696	0.639	0.186	0.334	0.472
Ours	0.414	8.701	2.651	0.423	0.655	0.883	0.205	0.411	0.540

❑ Ablation Studies

point	image	normal	CD↓	JSD	3D↓	F1↑	Rasterization setting	CD ↓	JSD-3D ↓	F1 ↑
✓			0.606	0.703	0.725		Coarse/smoothed	0.564	0.669	0.757
✓	✓		0.564	0.669	0.757		Fine/sharp	0.648	0.671	0.767
✓		✓	0.534	0.654	0.817					
✓	✓	✓	0.414	0.655	0.883					

5. Conclusions

- YUPU provides standardized acquisition-derived ALS point upsampling dataset.
- GCDM combines point, multi-view appearance, and normal conditions.
- Results demonstrate improved geometric fidelity and scene completion.

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